

Equation Reference and Verification

Complete listing of every equation used in `motion_planning_double_integrator.md` and in the benchmark harness, cross-referenced to the code and to the numerical checks in `verify_math.py`. Status column: **PASS** = verified to the residual shown by `python3 verify_math.py` (14/14 pass).

Notation: Δt time step, N horizon, n axes (joints/DOF), q position, $\dot{q}$ velocity, $\ddot{q}$ acceleration, u control.

1. Backbone dynamics

1.1 Double integrator (manipulator; paper §II.A)

Continuous, per axis:

$$\dot{x} = A_c x + B_c u, \quad x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, \quad A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad u = \ddot{q}. \quad (1)$$

A_c is nilpotent ($A_c^2 = 0$), so the matrix exponential truncates and the **exact ZOH** discretization is (paper §II.B):

$$e^{A_c \Delta t} = I + A_c \Delta t, \quad A_d = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad B_d = \int_0^{\Delta t} e^{A_c \tau} B_c d\tau = \begin{bmatrix} \frac{\Delta t^2}{2} \\ \Delta t \end{bmatrix}. \quad (2)$$

Code: `di_matrices`, step update in `plan_di`. Check: **E1** (residual 1.7×10^{-18}).

1.2 Triple integrator (AV lateral channel; `di_lateral.py`)

$$z = \begin{bmatrix} y \\ \dot{y} \\ \ddot{y} \end{bmatrix}, \quad A_c = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad u = \ddot{y} \text{ (jerk)}, \quad A_c^3 = 0. \quad (3)$$

$$A_d = \begin{bmatrix} 1 & \Delta t & \frac{\Delta t^2}{2} \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{bmatrix}, \quad B_d = \begin{bmatrix} \frac{\Delta t^3}{6} \\ \frac{\Delta t^2}{2} \\ \Delta t \end{bmatrix}. \quad (4)$$

Code: `_tri_matrices`, step in `plan_lateral_jerk`. Check: **E2** (4.6×10^{-16}).

2. Finite-horizon prediction

State rollout: $x_k = A_d^k x_0 + \sum_{j=0}^{k-1} A_d^{k-1-j} B_d u_j$, stacked over $k = 1..N$ (paper §II.B):

$$X = \Phi x_0 + \Gamma U, \quad \Phi = \begin{bmatrix} A_d \\ A_d^2 \\ \vdots \\ A_d^N \end{bmatrix}, \quad \Gamma = \begin{bmatrix} B_d & & \\ A_d B_d & B_d & \\ \vdots & & \ddots \\ A_d^{N-1} B_d & \dots & B_d \end{bmatrix}. \quad (5)$$

2.1 Double-integrator entries (split into position/velocity rows)

With $m = k - 1 - j$:

$$\Phi_p[k] = \begin{bmatrix} 1 & k\Delta t \end{bmatrix}, \quad \Phi_v[k] = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad \Gamma_p[k, j] = \Delta t^2 \left(m + \frac{1}{2}\right), \quad \Gamma_v[k, j] = \Delta t. \quad (6)$$

Code: `di_matrices`. Check: **E3** (4.2×10^{-17} , vs brute-force rollout).

2.2 Triple-integrator entries

$$\Gamma_p[k, j] = \Delta t^3 \left(\frac{1}{6} + \frac{m}{2} + \frac{m^2}{2}\right), \quad \Gamma_v[k, j] = \Delta t^2 \left(m + \frac{1}{2}\right), \quad \Gamma_a[k, j] = \Delta t, \quad (7)$$

$$\Phi_p[k] = \begin{bmatrix} 1 & k\Delta t & \frac{(k\Delta t)^2}{2} \end{bmatrix}. \quad (8)$$

Code: `_tri_matrices`. Check: **E4** (2.2×10^{-16}).

3. Receding-horizon QP (paper §III)

Decision variable $U = [u_0, \dots, u_{N-1}]$. Objective (paper §III.B):

$$\min_U \frac{1}{2} U^\top H U + h^\top U, \quad H = \Gamma^\top \bar{Q} \Gamma + \bar{R}, \quad h = \Gamma^\top \bar{Q} x_{\text{free, err}}, \quad (9)$$

$$x_{\text{free, err}} = \Phi x_0 - \mathbf{1}_N \otimes x_{\text{goal}}. \quad (10)$$

Implemented per axis with explicit position/velocity weights W_q, W_v (terminal multiplier γ on the last step) and control weight w_r :

$$H = \Gamma_p^\top W_q \Gamma_p + \Gamma_v^\top W_v \Gamma_v + w_r I, \quad (11)$$

$$h = \Gamma_p^\top W_q(\Phi_p x_0 - q_g \mathbf{1}) + \Gamma_v^\top W_v(\Phi_v x_0 - v_g \mathbf{1}). \quad (12)$$

This minimizes the explicit weighted least-squares cost

$J(U) = \sum_k W_{q,k}(q_k - q_g)^2 + W_{v,k}(\dot{q}_k - v_g)^2 + w_r \sum_k u_k^2$ (the OSQP $\frac{1}{2}H$ convention scales H, h by $\frac{1}{2}$ vs J ; the minimizer is identical). $H \succ 0$ since $w_r > 0 \Rightarrow$ strictly convex.

Code: `JointQP`. Check: **E5** ($\nabla J = 2HU + 2h$ matches, 4.4×10^{-16}).

3.1 Hard kinematic constraints (paper §III.C)

For $k = 1..N$ (velocity) and $k = 0..N - 1$ (acceleration), affine in U :

$$-\dot{q}_{\max} - \Phi_v x_0 \leq \Gamma_v U \leq \dot{q}_{\max} - \Phi_v x_0, \quad (13)$$

$$-u_{\max} \leq U \leq u_{\max}, \quad (14)$$

optional $-q_{\max} - \Phi_p x_0 \leq \Gamma_p U \leq q_{\max} - \Phi_p x_0$. Code: `JointQP.solve`.

3.2 Triple-integrator (AV) constraints (`di_lateral.py`)

$$|\ddot{y}| \leq a_{\text{lat,max}} \text{ (lateral accel), } |\ddot{y}| \leq j_{\max} \text{ (lateral jerk), } |\dot{y}| \leq \dot{y}_{\max}. \quad (15)$$

3.3 Exact state advance (ZOH, one step)

$$q_{k+1} = q_k + \Delta t \dot{q}_k + \frac{1}{2} \Delta t^2 u_k, \quad \dot{q}_{k+1} = \dot{q}_k + \Delta t u_k, \quad (16)$$

and the triple-integrator analogue

$$y_{k+1} = y_k + \Delta t \dot{y}_k + \frac{1}{2} \Delta t^2 \ddot{y}_k + \frac{\Delta t^3}{6} \ddot{\ddot{y}}_k, \text{ etc.}$$

Checks: **E10** (double, residual 0), **E11** (triple, residual 0).

4. Obstacle constraints (paper §IV; not exercised by the planning benchmarks)

Linearized polytope (hard, §IV.A): with translational Jacobian $J_v(q_0)$ and

$$p(k) \approx p_0 + J_v(q_0) \Delta q(k),$$

$$A_{\text{obs}} J_v(q_0) \Delta q(k) \leq b_{\text{obs}} - A_{\text{obs}} p_0. \quad (17)$$

APF target deflection (soft, §IV.B):

$$U_{\text{rep}}(p) = \frac{1}{2}\eta \left(\frac{1}{\|p - p_{\text{obs}}\|} - \frac{1}{\rho_0} \right)^2, \quad \|p - p_{\text{obs}}\| \leq \rho_0; \quad \delta q_{\text{obs}} = J_v^+(q) F_{\text{rep}}, \quad F_{\text{rep}} = -\nabla U_{\text{rep}}. \quad (18)$$

5. Path-first reference (TOTG / TOPP; `di_totg.py`)

5.1 Circular-blend geometry (Kunz–Stilman [10])

For interior waypoint W_i with unit leg directions u_1, u_2 :

$$\varphi = \arccos(u_1 \cdot u_2), \quad r = \frac{\delta_{\text{max}}}{\sec(\varphi/2) - 1}, \quad \ell = r \tan(\varphi/2), \quad (19)$$

(ℓ capped to $\leq \frac{1}{2} \min(L_1, L_2)$), then $r = \ell / \tan(\varphi/2)$). Tangent points and arc center:

$$P_s = W_i - \ell u_1, \quad P_e = W_i + \ell u_2, \quad C = W_i + r \sec(\varphi/2) \frac{u_2 - u_1}{\|u_2 - u_1\|}. \quad (20)$$

Max deviation from the corner: $\delta = r(\sec(\varphi/2) - 1)$. Check: **E8** (closest approach = 0.100 = δ_{max}).

5.2 Arc-length parameterization

With orthonormal $e_1 = (P_s - C)/r$, $e_2 \perp e_1$ toward P_e , and arc angle $a = (s - s_0)/r \in [0, \varphi]$:

$$q(s) = C + r(\cos a e_1 + \sin a e_2), \quad q'(s) = -\sin a e_1 + \cos a e_2, \quad q''(s) = -\frac{1}{r}(\cos a e_1 + \sin a e_2). \quad (21)$$

On straight segments $q(s) = P_0 + (s - s_0)d$, $q' = d$, $q'' = 0$. Arc length $L_{\text{arc}} = r\varphi$. Thus $\|q'(s)\| = 1$ everywhere (arc-length), and $\|q''(s)\| = 0$ on lines, $1/r$ on arcs — **discontinuous at seams**. Checks: **E6** ($\|q'\| = 1$, 2.2×10^{-16}), **E7** (C^1 tangent + curvature jump $0 \rightarrow 1/r$).

5.3 Velocity/acceleration along the path

$$\dot{q} = q'(s) \dot{s}, \quad \ddot{q} = q'(s) \ddot{s} + q''(s) \dot{s}^2. \quad (22)$$

The second (centripetal) term carries the q'' discontinuity and scales with $\dot{s}^2$.

5.4 Time-optimal parameterization (numerical-integration TOPP)

Let $x = \dot{s}^2$. Maximum-velocity curve from velocity limits:

$$\dot{s}_{\max}(s) = \min_j \frac{\dot{q}_{\max,j}}{|q'_j(s)|}, \quad x_{\text{vel}} = \dot{s}_{\max}^2. \quad (23)$$

Acceleration limits give, per axis, an admissible $\ddot{s}$ interval (affine in x):

$$-a_{\max,j} \leq q'_j(s) \ddot{s} + q''_j(s) x \leq a_{\max,j} \Rightarrow \ddot{s} \in [L(s, x), U(s, x)], \quad (24)$$

and the accel-MVC is the largest x with $L \leq U$ (found by bisection). Forward/backward sweep with $x' = dx/ds = 2\dot{s}$:

$$x_{k+1}^f = \min(x_k^f + 2U\Delta s, \text{MVC}_{k+1}), \quad x_{k-1}^b = \min(x_k^b - 2L\Delta s, \text{MVC}_{k-1}), \quad (25)$$

$$x = \min(x^f, x^b, \text{MVC}). \quad (26)$$

Code: `topp`, `_u_interval`. Checks: **E9** (path-domain $|\dot{q}| \leq \dot{q}_{\max}$, residual 0), **E9b** (admissible $\ddot{s}$ interval exists at every s).

5.5 The $s \rightarrow t$ conversion (where it breaks)

$$\ddot{s} = \frac{1}{2} \frac{dx}{ds}, \quad \Delta t_k = \frac{2\Delta s}{\dot{s}_k + \dot{s}_{k+1}}, \quad t(s) = \sum_k \Delta t_k. \quad (27)$$

$$\boxed{t(s) = \int \frac{ds}{\dot{s}} \text{ is singular as } \dot{s} \rightarrow 0} \Rightarrow \text{near-reversal } (r \rightarrow 0) \Rightarrow T \rightarrow \infty \text{ (B3 STAI)} \quad (28)$$

Reconstruction at uniform time: $s(t) = \text{interp}$, then $q = q(s(t))$ and $\ddot{q}$ from (22) — the q'' jump makes $\ddot{q}$ discontinuous and can exceed $a_{\max}$.

Code: `plan_totg`. Check: **E13** ($\ddot{s} = \frac{1}{2} dx/ds, 6 \times 10^{-5}$).

6. Autonomous-vehicle kinematics (steering; `benchmark_av.py`)

Kinematic bicycle, wheelbase L , planar velocity $v = (\dot{x}, \dot{y})$, accel $(\ddot{x}, \ddot{y})$:

$$\text{speed} = \|v\|, \quad \kappa = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\|v\|^3}, \quad \delta = \text{atan}(L\kappa), \quad a_{\text{lat}} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\|v\|} = \|v\|^2 \kappa. \quad (29)$$

Steering rate $\dot{\delta} = d\delta/dt$. Code: `steering`. Check:

E12 (constant-curvature circle: $\kappa = 1/R$, $\delta = \text{atan}(L/R)$, residual 5.6×10^{-17}).

6.1 Integrator-order principle (AV)

At cruise speed V ($x = Vt$, $\dot{y} \ll V$): $\kappa \approx \ddot{y}/V^2$, so

$$\delta \approx \text{atan}(L\ddot{y}/V^2), \quad \dot{\delta} \approx \frac{L}{V^2} \ddot{\dot{y}} \text{ (small angle)}. \quad (30)$$

Bounding lateral jerk $|\ddot{\dot{y}}| \leq j_{\text{max}}$ (15) therefore bounds steering rate by construction — motivating the **triple** integrator (one order above the manipulator's double integrator).

7. Benchmark metrics

$$\text{accel ratio} = \frac{\|\ddot{q}\|_{\infty}}{\ddot{q}_{\text{max}}} (> 1 \Rightarrow \text{limit broken}), \quad \text{jerk}_k = \frac{\ddot{q}_{k+1} - \ddot{q}_k}{\Delta t}, \quad (31)$$

$$\text{accel jump} = \max_k \|\ddot{q}_{k+1} - \ddot{q}_k\|_{\infty}, \quad \text{conditioning} = \max_s \frac{1}{\dot{s}(s)} \text{ (TOTG only)}. \quad (32)$$

8. Verification summary

python3 `verify_math.py` → **14/14 PASS**:

Check	Equation(s)	Residual
E1	(2) double-int ZOH	1.7e-18
E2	(4) triple-int ZOH	4.6e-16
E3	(6) DI prediction Φ, Γ	4.2e-17
E4	(7,8) triple-int prediction	2.2e-16
E5	(9–12) QP cost algebra	4.4e-16

Check	Equation(s)	Residual
E6	(21) endpoints + $ q' = 1$	2.2e-16
E7	(21) C^1 + curvature jump	1.4e-05
E8	(19,20) blend deviation $\leq \delta_{\max}$	0.10 (=cap)
E9 / E9b	(23–26) TOPP feasibility	0 / 0
E10	(16) DI time consistency	0
E11	(16) triple-int time consistency	0
E12	(29) steering formula	5.6e-17
E13	(27) $\ddot{s} = \frac{1}{2} dx/ds$	6.0e-05

Non-machine-precision residuals (E7, E8, E13) are finite-difference/grid artifacts of the *check itself*, not of the underlying equation, and are within the stated tolerances.